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Information & Communication Technology

OBJECTIVES

- Understand microprocessor-controlled devices:
  - Positive and negative effects on everyday life in the home.
  - Positive and negative effects on monitoring and controlling transport.
- Learn about health issues from computer use:
  - Common problems like RSI, back/neck strain, eye strain, and headaches.
  - Causes of these health issues and ways to prevent them.

5.1

Microprocessor-controlled devices

Microprocessor Controlled Devices

- ❖ Devices enhance convenience but have both positive and negative impacts.

Labour-saving devices

- ➢ Washing machines, robotic vacuums, smart fridges.

Functional devices

- ➢ Alarm clocks, mobile phones, entertainment systems.

Lifestyle, Leisure Time, and Physical Fitness

Advantages

- ➢ Eliminates manual tasks.
- ➢ Provides more leisure time.
- ➢ Enables remote control of devices via smartphones.
- ➢ Enhances security (e.g., automated alarms).
- ➢ Smart fridges promote healthy lifestyles and reduce food waste.

Disadvantages

- ➢ Promotes unhealthy lifestyles due to lack of exercise.
- ➢ Dependency on devices leads to laziness and loss of skills.
- ➢ Cybersecurity threats from internet-connected devices.

Effects of Microprocessor Controlled Devices

Advantages

- ➢ Saves energy with efficient auto-shutdown features.
- ➢ Simplifies programming through QR codes.

Disadvantages

- ➢ Non-repairable devices contribute to waste.
- ➢ Intimidates technophobes.
- ➢ Devices left on standby waste electricity.

Data Security Issues

- ❖ Internet-connected devices are prone to hacking.

Risks

- ➢ Hackers accessing personal data via smart devices.
- ➢ Stolen credit card info from internet-connected fridges.
- ➢ Finding holiday dates through hacked heating controllers.

Mitigation

- ➢ Use strong passwords, update software regularly.

Social Interactions

Positive Impacts

- Easier to make friends and connect with similar interests.
- Cost-effective communication (VoIP, video calls).

Negative Impacts

- Reduces face-to-face interaction, causing isolation.
- Online communication encourages rudeness and cyberbullying.
- Increases anxiety in real-life interactions.

Microprocessors in Transport

Applications

- ➢ Traffic monitoring and control (traffic lights, ANPR).
- ➢ Air traffic control systems.
- ➢ Railway signaling systems.

Smart Road Systems and Signs

- ❖ Smart motorways
  - ➢ Central computer systems monitor traffic and manage motorway signs.
  - ➢ Adjust traffic flow during accidents or congestion to reduce delays.
- ❖ City traffic lights: Computer-controlled for efficiency.
- ❖ Security risks: Hacking could disrupt road networks with significant safety implications.

Rail and Airline Network Control Systems

Computerized systems

- ❖ Coordination of trains and planes for safer, efficient timetables.
- ❖ Reduction of human error in rail and air accidents.

Advantages

- ➢ Reduced traffic jams with adaptable smart motorways.
- ➢ Efficient transport networks with automated penalties for traffic offenses.
- ➢ Enhanced safety with minimized human error.

Disadvantages

- ➢ Vulnerable to hacking, causing disruption.
- ➢ System failures could halt transport systems.
- ➢ Privacy concerns with movement tracking via ANPR.

Autonomous Vehicles

- Use sensors, cameras, microprocessors, and actuators for tasks like braking, steering, and parking.
- Build 3D surroundings using radar and ultrasonic sensors.
- Examples: Cars, buses, and trains.

Autonomous Vehicles

Advantages

- ➢ Safer due to the removal of human error.
- ➢ Environmentally friendly and traffic-efficient:
- ➢ Reduced congestion and optimized lane capacity.
- ➢ Improved braking and acceleration for smoother traffic flow.
- ➢ Stress-free parking and shorter travel times.

Disadvantages

- ➢ High initial costs and maintenance requirements.
- ➢ Vulnerable to hacking:
- ➢ Potential for data breaches or system manipulation.
- ➢ Issues with sensor disruptions in extreme weather.
- ➢ Passenger skepticism and potential unemployment for taxi drivers.

Security and Safety Concerns

- ❖ Autonomous systems rely on software updates and secure sensor data.
- ❖ Hackers could block sensors or feed false data.
- ❖ Hackers could access personal information or monitor vehicle location.

If resolved, autonomous transport could eliminate many safety issues for occupants and pedestrians.

Autonomous Trains

- ❖ Uses sensors, cameras, LiDAR, GPS, and actuators for operation and safety.
- ❖ LiDAR creates a 3D image of surroundings, while cameras and sensors manage door and platform safety.

Advantages

- ➢ Improved punctuality and safety by removing human error.
- ➢ Reduced energy consumption and running costs.
- ➢ Increased frequency of trains during peak times.

Disadvantages

- ➢ Vulnerable to hacking and high initial costs.
- ➢ Challenges with passenger behavior and technology acceptance.

Pilotless Aeroplanes

- ❖ Sensors detect turbulence, cabin depressurization, and conduct self-testing.
- ❖ GPS and actuators manage navigation, speed, and flight controls.

Advantages

- ➢ Increased passenger comfort and reduced running costs.
- ➢ Improved safety by eliminating pilot-induced errors.
- ➢ Better aerodynamics without a cockpit.

Disadvantages

- ➢ Security concerns (e.g., hacking or handling emergencies).
- ➢ Software glitches and passenger reluctance to adopt.

5.2

Potential health problems related to the prolonged use of IT equipment

Categories

Back and Neck Strain

- **Causes**: Poor posture and prolonged sitting.
- **Solutions**: Adjustable chairs, footrests, tiltable screens.

Repetitive Strain Injury (RSI)

- **Causes**: Continuous keyboard/mouse use.
- **Solutions**: Ergonomic keyboards, wrist rests, voice-activated software, regular breaks.

Eyestrain

- **Causes**: Long screen exposure, improper lighting.
- **Solutions**: Anti-glare screens, regular breaks, proper lighting, eye tests.

Categories

Headaches

- **Causes**: Reflections, flickering screens.
- **Solutions**: Anti-glare screens, regular breaks, eye tests.

Ozone Irritation

- **Causes**: Emissions from laser printers.
- **Solutions**: Ventilation, designated printer rooms, switch to inkjet printers.